Network Topologies

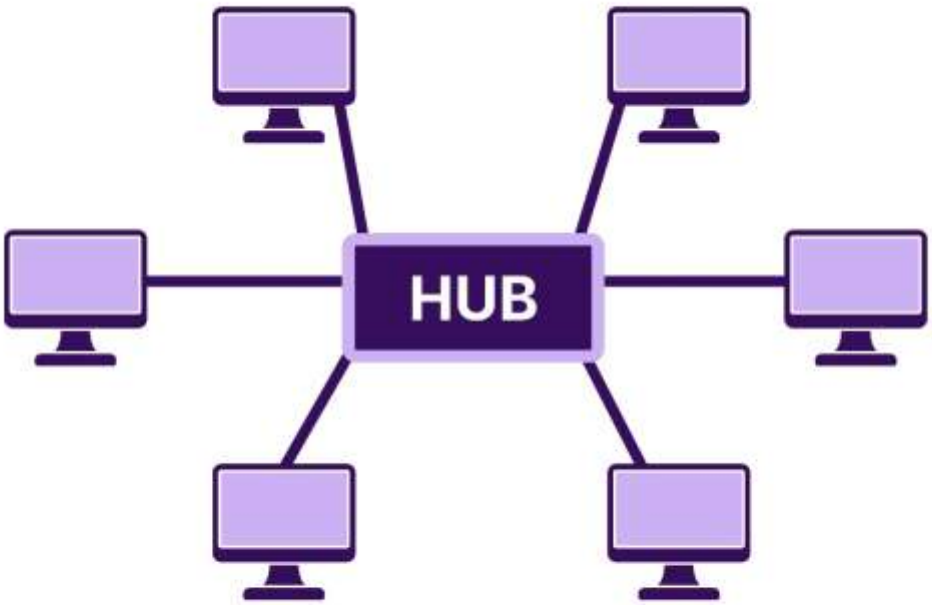
Tasks

1.Draw and paper label diagrams of star, bus, and mesh network topologies on or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.

ANS:

Network topology means the arrangement or layout of computers and devices connected in a network.

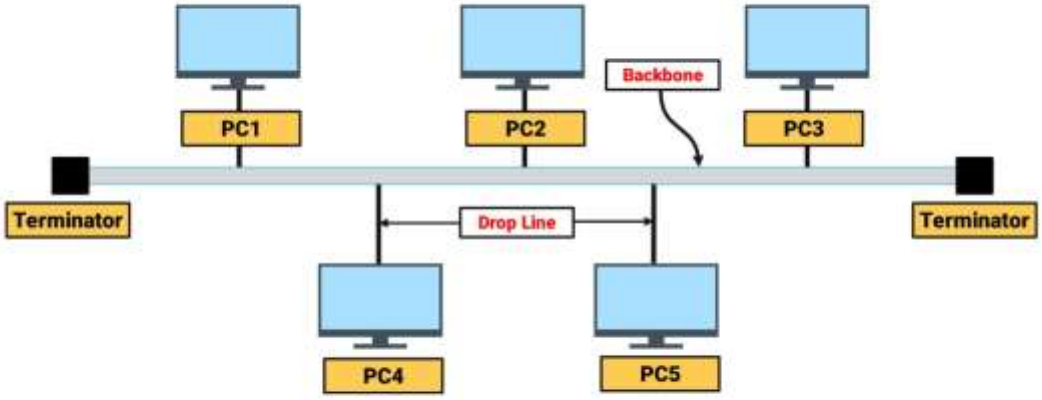
1.Star Topology:



Real-world example: Office Wi-Fi / Computer Lab

Explanation: In an office or computer lab, all computers connect to a central switch. If one computer stops working, the other computers can still communicate.

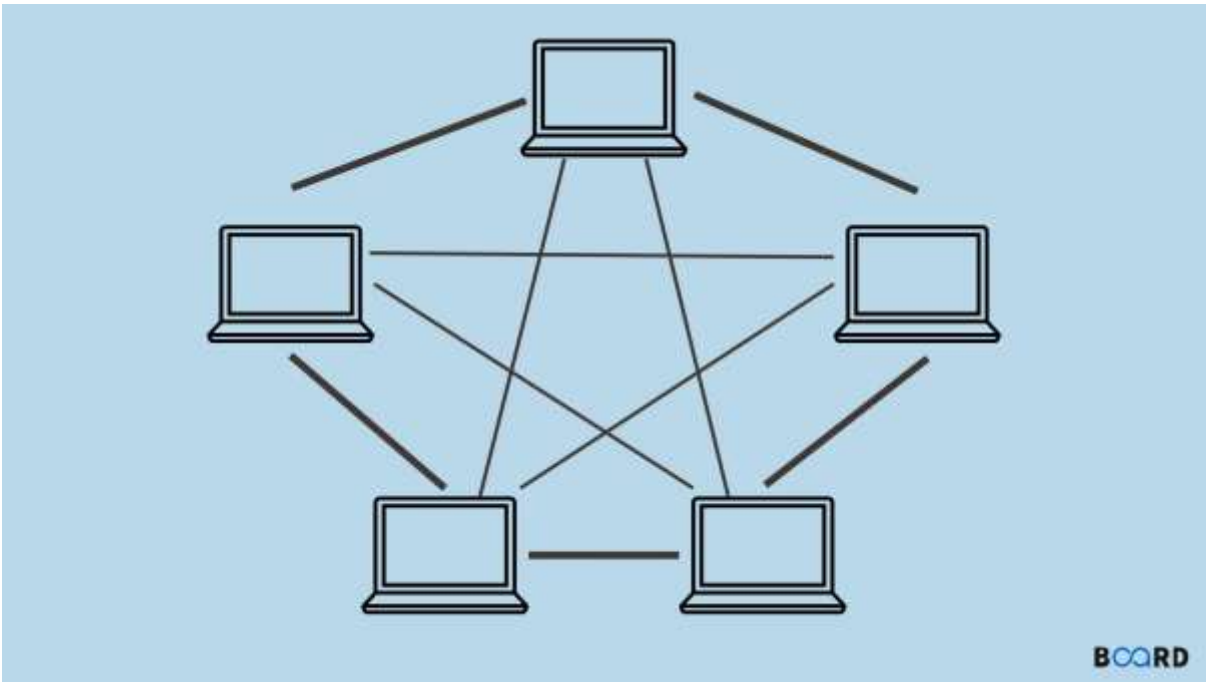
2. Bus Topology:



Real-world example: Small old office LAN

Explanation: In a bus topology, all computers share one main cable to communicate. It is simple and uses less cable, but if the main cable fails, the entire network stops working.

3. Mesh Topology:



Real-world example: Internet backbone / Data Centre Network

Explanation: In mesh topology, devices have multiple connections with each other. If one connection fails, data can travel through another path, making the network more reliable.

Topology	Structure	Main advantage
Star	Central switch	Easy to manage
Bus	Single main cable	Less cabling
Mesh	Multiple connections	High reliability

2.Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

ANS:

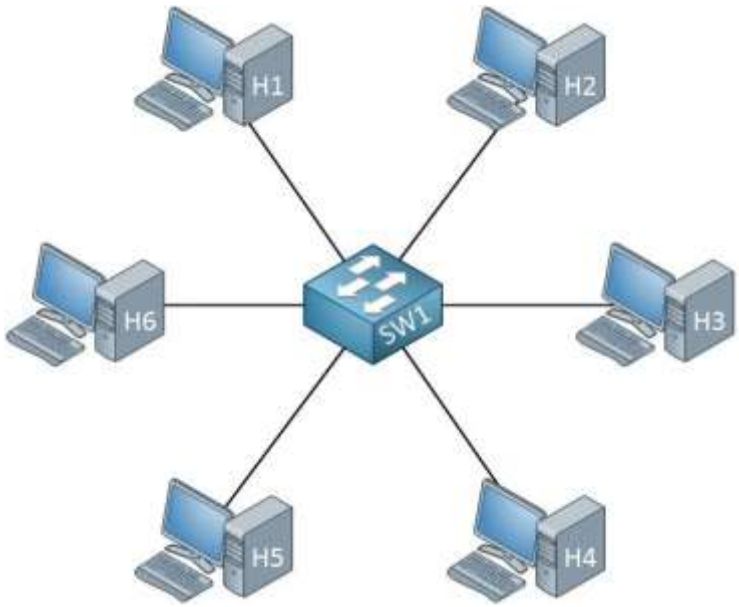
Comparison of Network Topologies

Network topology describes how computers and devices are connected in a network. There are four common types: Star, Bus, Ring, and Mesh.

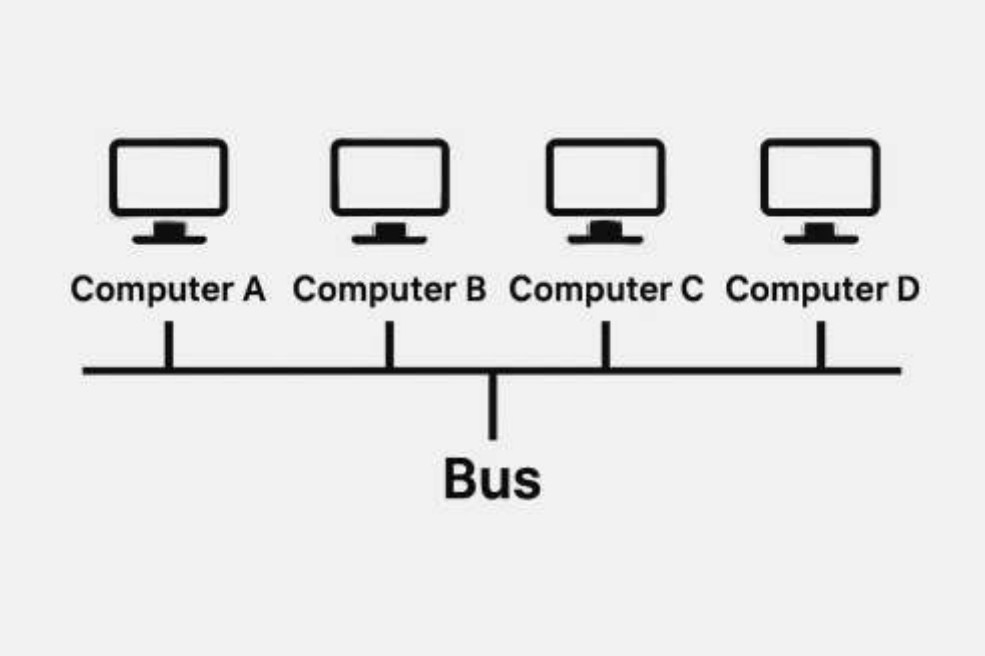
Topology	Advantages (Pros)	Disadvantages (Cons)	Best Use Case
Star	1. Easy to install. 2. Easy to manage. 3. Failure of one computer does not affect others. 4. Easy to detect network problems.	1. Depends on central switch. 2. Requires more cables. 3. Installation cost is higher. 4. If the central switch fails, the whole network stops.	Offices, schools and computer labs. Example: Instagram – its servers use interconnected network infrastructure to handle user requests.
Bus	1. Easy to set up. 2. Requires less cable. 3. Low installation cost. 4. Suitable for small networks.	1. Main cable failure stops the network. 2. Difficult to troubleshoot. 3. Performance decreases with more devices.	Small or older LAN networks. Example: Zomato – a simple bus network could connect devices in a small restaurant.

		4. Limited cable length.	
Ring	1. Data travels in an organized manner. 2. No data collision in token-based ring networks. 3. Equal access for connected devices. 4. Predictable data transmission.	1. One connection failure can disrupt the ring. 2. Adding or removing devices can be difficult. 3. Troubleshooting can be complex. 4. Performance can decrease if a device or link has problems.	Industrial networks and some older LAN systems. Example: Paytm – a ring-style network could connect devices in a payment-processing environment.
Mesh	1. Highly reliable. 2. Multiple paths for data transmission. 3. Failure of one link does not stop the entire network. 4. Better fault tolerance and availability.	1. Expensive to install. 2. Requires more cables and connections. 3. Complex configuration. 4. Difficult to maintain in large networks.	Data centres, wireless mesh networks and critical communication systems. Example: WhatsApp – its communication infrastructure can use multiple network paths for reliable data delivery.

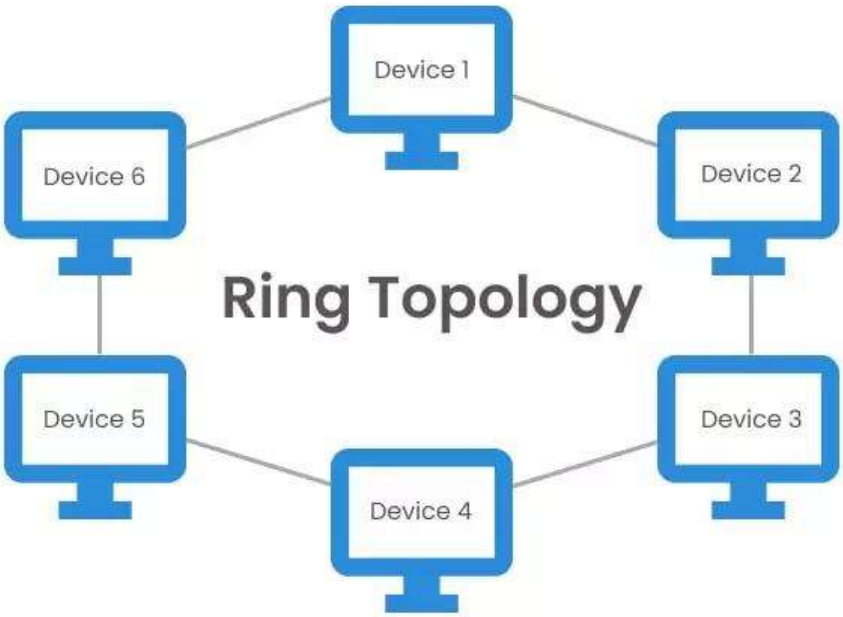
1. Star Topology



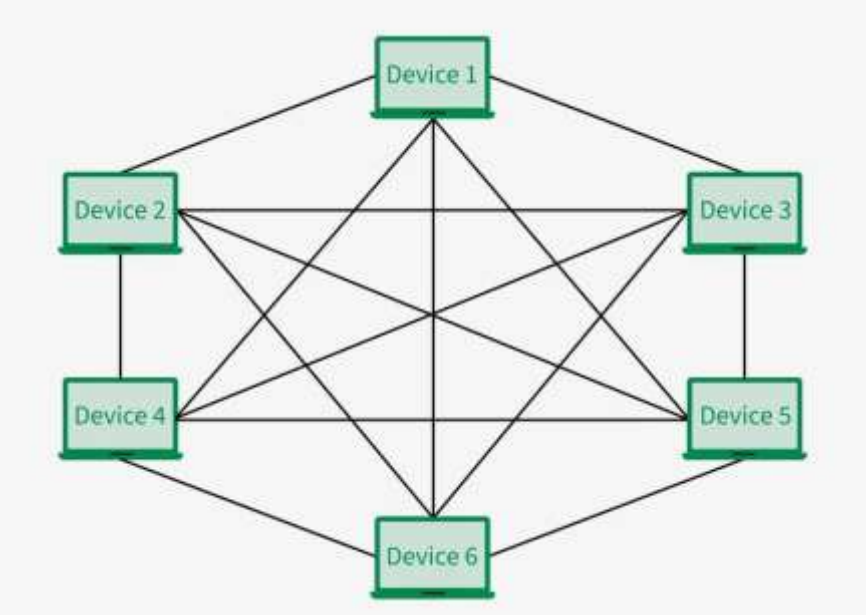
2. Bus Topology



3. Ring Topology



4. Mesh Topology



1. Star: Easy to manage.
2. Bus: Cheap and simple.
3. Ring: Sends data in order.
4. Mesh: Very strong and reliable network.

3.Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.

ANS:

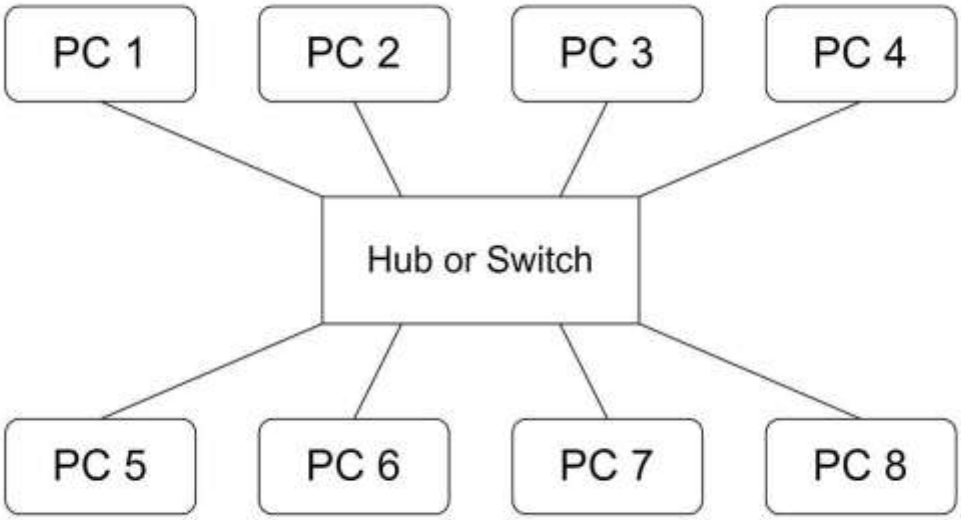
Chosen Topology: Star Topology

1. I would choose Star Topology for a game like PUBG or Free Fire.
2. All players connect to one main game server.
3. It helps players play smoothly and share data quickly.
4. If one player disconnects, other players can continue playing. It costs more because a main server is needed.

**4.Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines.

Hint: Consider factors like cost, ease of troubleshooting, and future expansion.**

ANS:



1. I would choose Star Topology for a small warehouse.
2. All 8 computers are connected to one main switch.
3. It is easy to find problems and fix them.
4. If we buy more computers in the future, we can connect them easily.

Real-World Example: Flipkart warehouse – Employees use connected computers to check customer orders, update stock and manage deliveries.